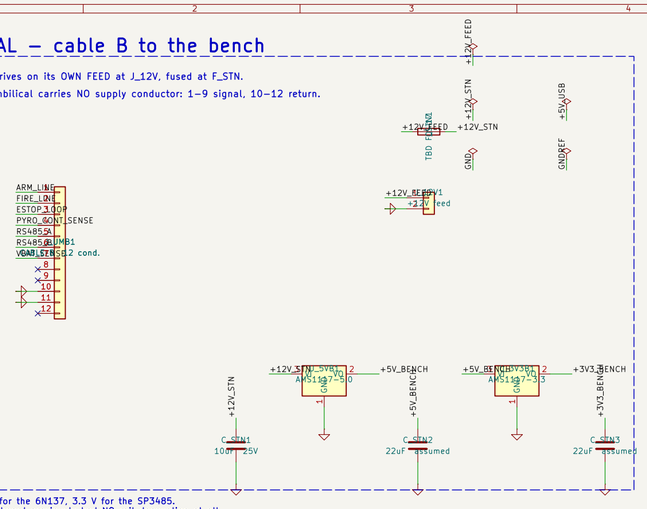
## 1. UMBILICAL — cable B to the bench

The station supply arrives on its OWN FEED at J_12V, fused at F_STN.

The 12-conductor umbilical carries NO supply conductor: 1–9 signal, 10–12 return.




Bench-side rails: 5 V for the 6N137, 3.3 V for the SP3485.
C_STN1–4 added: this board previously had NO rail decoupling at all.
22 uF values are ASSUMED from the AMS1117 stability note — confirm (O-20).

## 2. CONTROLS — mechanical, panel-mounted

All four are PANEL controls wired to these terminals, not PCB parts.




S_ARM: removable key, carried by the ignition officer. Closed = bench relay K1 pulls in.

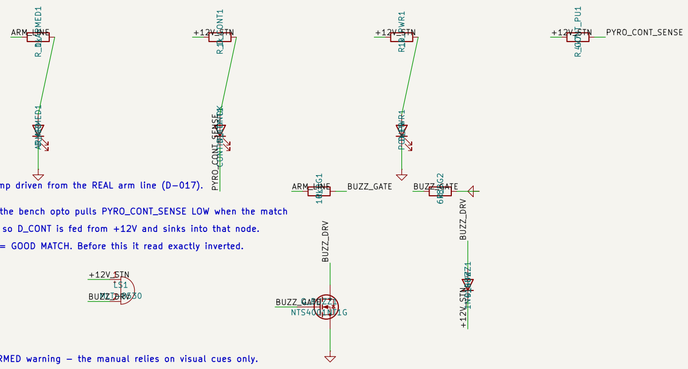
S_FIRE: guarded panel momentary, NO.
Fed FROM ARM_LINE, so FIRE is dead unless ARMED.
SKRPABE010 is NOT acceptable here (D-044).

S_ESTOP: latching mushroom, NORMALLY CLOSED.
Opening it drops bench master relay K4.
NC, so a broken wire also stops the bench.

H-2 FIX: the CONTINUITY test button is REMOVED.
As drawn it shorted PYRO_CONT_SENSE to ground, which is the lamp-good state, so the test reported a good match whether or not one was connected. A test that cannot fail is worse than no test.
Sensing is continuous and needs no button to initiate it.

OPEN DESIGN QUESTION: because sensing is continuous, current flows through the e-match whenever the bench is powered. Gating that needs a decision — see the audit, H-2.

## 3. INDICATORS & AUDIO




ARMED lamp driven from the REAL arm line (D-017).

H-1 FIX: the bench opto pulls PYRO_CONT_SENSE LOW when the match conducts, so D_CONT is fed from +12V and sinks into that node.
LAMP LIT = GOOD MATCH. Before this it read exactly inverted.

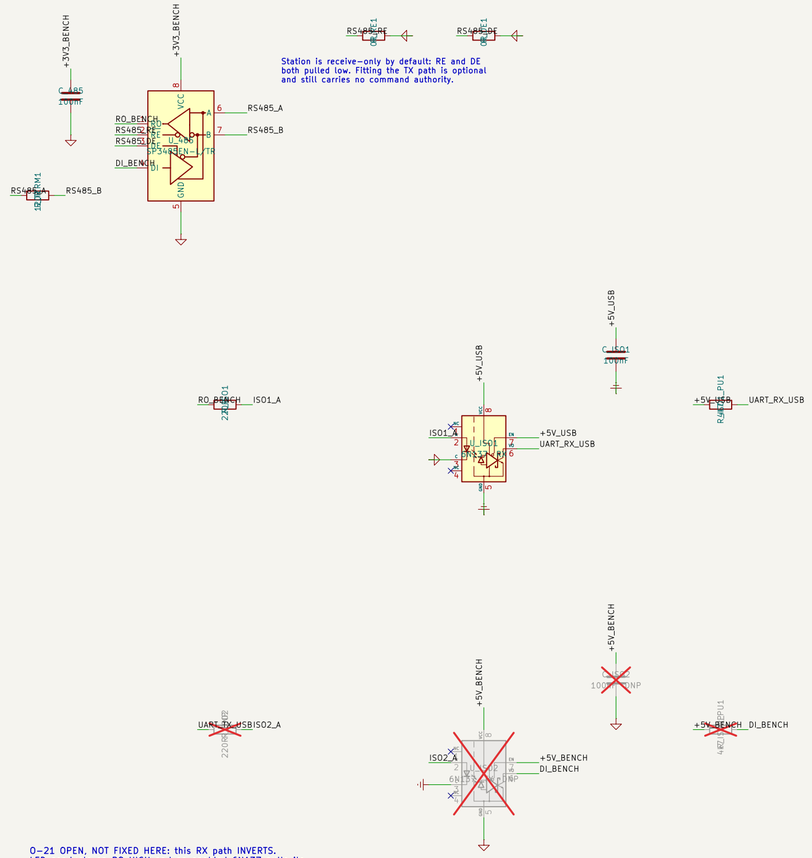
Audible ARMED warning — the manual relies on visual cues only.

## 4. TELEMETRY — RS-485 in, isolated, out to USB

ISOLATION BARRIER down the middle of this block.

LEFT = bench-referenced (GND). RIGHT = laptop side (GNDREF).

No isolated DC-DC: the laptop's USB supplies the isolated side (D-056).

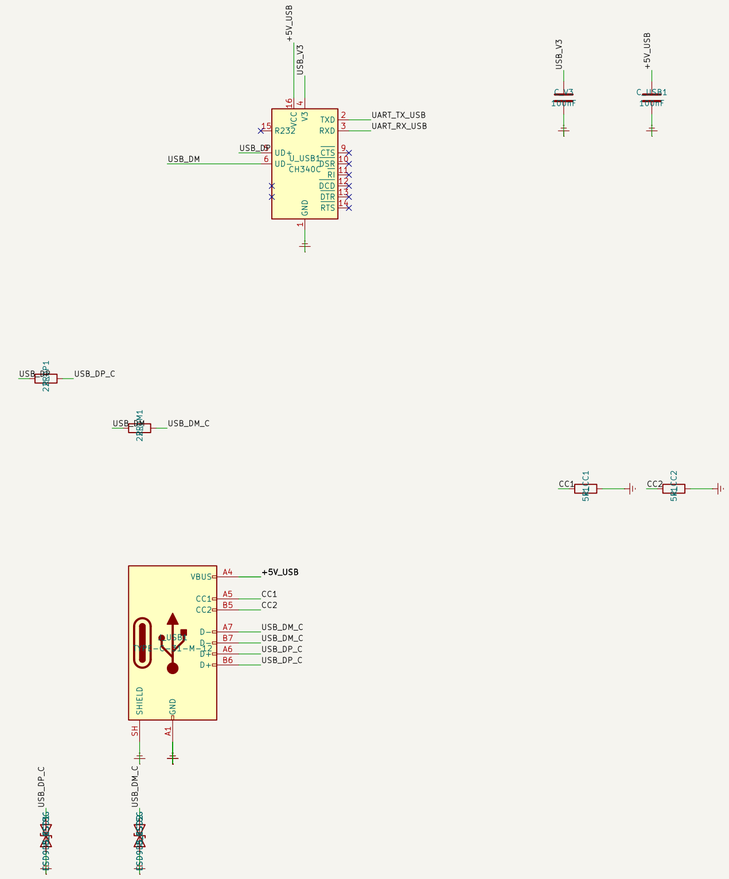



Station is receive-only by default: RE and DE both pulled low. Fitting the TX path is optional and still carries no command authority.

O-21 OPEN, NOT FIXED HERE: this RX path INVERTS.
LED conducts on RO HIGH and an enabled 6N137 pulls its output LOW, so an idle mark reaches the CH340C as a continuous break. Telemetry cannot work as drawn. The fix is a part change, not a rewire — see REV-A1-CHANGES.

6N137 is fast enough for 115200 baud. The TLP521 / PC817 already in stock are NOT — do not substitute (O-6).

## 5. USB-C — laptop side, GNDREF domain




CC pull-downs 5k1 mark this as a UFP sink.
Data series 22R, one ESD diode per line.

## 6. MEASUREMENT OUTPUT — works with no software at all

PATH 2 IS DEFERRED FOR REV A. The SD log is the measurement.




J_DAQ IS NOT FITTED IN REV A (O-19).
It was wired to umbilical pins 8/9, which reach NOTHING on the bench: no buffer, no source. A connector present at both ends of a dead pair reads as a working analog backup, so it is unpopulated and its pins are flagged rather than left inviting a DAQ.
Pin 3 also exported bench GND to the laptop bench, which bridges the isolation barrier whenever a mains-powered DAQ shares it.
Loss of path-2 redundancy for Rev A is recorded in REQUIREMENTS.md.




Battery voltmeter: the pre-test >= 12 V check (manual p. 12), read without walking to the bench.

Review owner: Emilio Guadarrama. Generated by hardware/tools/gen_station.py
TWO GROUND DOMAINS: GND = bench-referenced, GNDREF = laptop/USB. Never joined.
PRELIMINARY — NOT APPROVED. ARM, FIRE and E-STOP are mechanical, panel-mounted.

**Ignitia Rocket Lab**

Sheet: /
File: talavera-station.kicad_sch

***Title: TALAVERA v1 — Control Station Board***

Size: A2 | Date: 2026-11-06 | **Rev: A**
KiCad E.D.A. 10.0.5 | Id: 1/1

TBD values await O-2 (load cell) and O-6 (isolator part). ARM / FIRE / E-STOP carry no firmware.